

FORT ORANGE DISTRICT

2018 CHAMPIONSHIP QUALIFIER

OFFICIAL PINEWOOD DERBY BUILD RULES

PRO STOCK SERIES

General:

1. Cars must be made from official BSA Official Grand Prix Pinewood Derby Kit.
2. Cars must be made for this racing season, no cars (or parts of cars such as wheels or axles) built prior to September 1, 2017 are allowed.
3. Cars arriving after official registration time ends will not be eligible to race and scouts must be present to race.
4. Those showing unsportsmanlike conduct will be eliminated from the competition. Attempts to circumvent the spirit of these rules may lead to disqualification as determined by Derby judges.

Length, Width and Clearance:

1. The maximum overall width (including wheels and axles) shall not exceed 2-3/4 (2.750) inches.
2. The maximum height of the vehicle including wheels should not exceed 3-1/2 inches.
3. The maximum overall length shall not exceed 7 inches.
4. The minimum width between the wheels shall be 1-3/4 (1.750) inches so the car will clear the center guide rails on the track.
5. The minimum clearance between the bottom of the car and the track surface shall be 3/8 (0.375) inch so the car will clear the center guide rails on the track.
6. The wheel-base must be at least 4 inches and must ensure that no portion of the wheels extend beyond the ends of the car body. The front two wheels of the car must be positioned directly across the body from one another and the rear two wheels of the car must be positioned directly across the body from one another. Axle holes may be drilled into the body of the car as long as all other specifications are met.
7. No part of the car or any attachment to the car may be capable of coming into contact with the track other than the wheels.

Weight and Appearance:

1. Weight shall not exceed **5.0 ounces**. The reading of the Official Race Scale will be considered final.
2. The car may be cut, carved, sculpted, and/or sanded in order to enhance performance and/or appearance.
3. The car may be hollowed out and built up to the maximum weight by the addition of solid materials only, provided the material is securely built into the body or firmly affixed to it. No liquids or loose materials of any kind are permitted in or on the car.
4. No weights or other objects may be taped to the car. Only permanently fastened weights and details are permitted.
5. Details such as steering wheel, driver, spoiler, decals, painting and interior details are permissible as long as these details do not exceed the maximum length, width or weight specifications.
6. Cars with wet paint, glass, excessively fragile parts, or energized electronic or lighting devices, will not be accepted.
7. The bottom surface of the car should be free of any sharp points or objects to protect the track and stopping pads.
8. No part of the car may protrude beyond the starting pin. The leading edge of the car must be at least 1/2 (0.50) inch wide at the center of the car to avoid being staged crooked, and must not be more than 1 inch above the wheel lane of the track. For design considerations, the center pin is approximately 1-3/4 (1.750) inches above the track surface.

Lubrication:

1. Only dry powdered graphite will be allowed for lubricating the wheels. Oils, silicone sprays, or any other lubricants that may foul the track surface are strictly prohibited.
2. A car may be lubricated before inspection. Lubrication will not be allowed after the car has been officially weighed in.

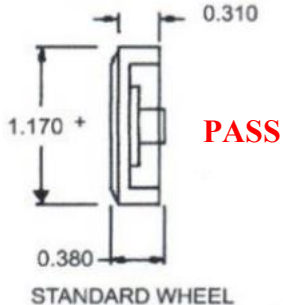
Wheels and Axles:

1. Wheels shall be only as provided in the BSA Official Grand Prix Pinewood Derby Kit, or BSA Official Grand Prix Pinewood Derby replacement wheels.
2. All molded wheel lettering and numbering, both inside and outside, must remain complete and be visible for inspection. The fluting and other BSA markings on the outside wheel area must remain visible for inspection.
3. Wheels may be lightly sanded to smooth out molding imperfections on the tread area. This light sanding is the only modification allowed. Outside diameter of wheel may not be reduced to less than 1.170 inches. Outer wheel surface must not be reshaped or have the contour changed in any way in an attempt to minimize tread contact or alter aerodynamics. Tread surface must be flat and parallel to the wheel bore.

4. Coning the hubs, truing the inside edge of the wheel, and removing the outer hub step down is allowed, as long as overall wheel width is not reduced below 0.36 inch.
5. No material may be removed from the inside wheel surfaces at any point.
6. The weight of the wheel shall not be increased or decreased except where limited amounts of material are removed from the tread or inside tread edge to remove minor imperfections. No material such as glue, fingernail polish, or tape, may be added to the inside of the wheel increasing its weight.
7. Judges reserve the right to measure the wheel with calipers to verify dimensional compliance with official wheel size restrictions.
8. Wheel bore treatment is allowed including polishing, sanding, and/or tapping. Wheel bores may not be filled and re-drilled to alter bore diameter or to achieve better fit with the axle (no material may be added to the wheels).
9. There must be at least four wheels on the car, however it is not required that all four wheels make contact with the track surface – one wheel may be lifted from the track surface. Each wheel must be mounted on an axle, on the outside of the car, on the side of the car. One wheel can have its tread surface be up to a 45-degree angle from the track surface. All other wheels must be vertical (i.e. the camber angle of at least 3 wheels must be 0 degrees). Each wheel must be attached by a compliant axle. Non-dominant / raised wheels must meet wheel specifications.
10. Nail type axles are to be used. The tip of the nail axle must be visible to allow inspectors to verify nail axles are being used, whether through ¼" inspection holes or open axle slots. Builders may or may not choose to use the axles found in the BSA Official Grand Prix Pinewood Derby Kit and replacement wheel sets.
11. Modifications to the axles are allowed that include straightening, sanding, polishing, canting for tracking purposes, grooving, beveling, and tapering of axle head as long as the diameter of each axle is not reduced to less than 0.084 inch and all other specifications are met.
12. Wheel bearings, washers, bushings, and hub caps are prohibited.
13. The car shall not ride on any type of springs.
14. The car must be free-wheeling, with no starting device or other type of propulsion such as adhesives, magnets, rubber bands, motors, jets, etc.

CLARIFICATION OF DISQUALIFYING / ALTERED WHEELS

No alteration, narrowing, lathe cutting, or re-shaping of wheels (inside or outside) is allowed!!!!

 PASS STANDARD WHEEL		FAIL	FAIL	FAIL	FAIL
					
		ROUNDED WHEEL	"V" WHEEL	CUPPED WHEEL	REDUCED DIAMETER
FAIL	FAIL	FAIL	FAIL	FAIL	FAIL
					
"H" WHEEL	RIBBED WHEEL	SPEED WHEEL	SLANTED OUT	SLANTED IN	KNIFE EDGE

NO LIGHTENING OF THE WHEEL IS ALLOWED

LIGHT SANDING OF WHEEL TREAD TO REMOVE IRREGULARITIES IS PERMISSABLE. WHEEL TREAD MUST BE FLAT!

MINIMUM WHEEL DIAMETER ALLOWABLE IS 1.170 INCHES.
MINIMUM WHEEL WEIGHT OF EACH WHEEL IS 2.45 GRAMS.

Please note: There are after-market modified wheels available that have been **lightened**. This is usually done by turning the wheels on a lathe and removing material from the inside of the wheel. These wheels are **NOT** allowed and can be easily recognized at inspection. Cars with third party manufactured or modified wheels will not be permitted to race.